1. Title

- TradeFlow AI. Track: Reinvent Commerce.
- Small cross-border sellers lose margin negotiating factory quotes alone, or pay an agent to do it.
- One operator. The engine sets the price. A human sends the offer.
- Prototype. No factory has closed a deal through this.
- Public repo: https://github.com/Fu93/tradeflow-ai. License: MIT.

2. Problem

- Cross-border buyers either negotiate alone, often with a language penalty, or pay an agent.
- TradeFlow is a prototype of a third arrangement: one operator, software doing the arithmetic, the person keeping the send button.
- It is not a deployed company and it has not bought from a real factory.

3. Market, revenue, competition

- Target user is a hypothesis: a small cross-border seller who sources from factories and negotiates in chat.
- HMRC 2025: 334,294 UK goods importers. SAM proxy 132,484 to 239,616 (formula in docs/MARKET_ANALYSIS.md).
- UK goods from China, 2025: GBP 69.7 billion (ONS/DBT). Includes precious metals and large corporates.
- Savings-pool ceiling GBP 1.05 billion to GBP 2.09 billion. Large-buyer 1.5 to 3 percent band. Not a TradeFlow result.
- Umbrellas (HS 6601) are the demo wedge. Other codes use the Comtrade preview script. Customs unit value is reference only and does not set the price.
- Revenue model, proposed and not earned: a desk subscription, or a share of savings versus the first-round best quote. No revenue is claimed.
- Public prices: Accio, and Alibaba RFQ free for buyers. Pactum, Keelvar, Fairmarkit, Jaggaer, Zycus, Globality are quote-based.

4. Split

- Hybrid desk. The criteria this workflow is written for are Hybrid, Becreatives, and Vibe Generation.
- Solver decides. The decision core does not ask a language model what to pay.
- Model phrases. An optional call can classify a message or draft wording.
- Human sends. The Chrome extension (Manifest V3) shows a card. It cannot press Send.
- Code: src/engine/solver.py, src/engine/multi_solver.py, extension/.

5. Decision core

- A belief about the supplier reserve, then a vectorised Expectimax search over openings and concession curves.
- Read the recommended price and the breakdown probability off the dashboard you just captured.
- This slide does not state a latency. Use the number on the screen.

6. AMD language layer

- Sample endpoint: https://developer.amd.com.cn/radeon/api/v1.
- Committed log results/real_llm_benchmark_100.json (2026-10-15): model Qwen3.8-Flash-Next, 348 calls, average latency 16.81 s, cost $0.108444.
- That file does not store an HTTP status. status: ok from scripts/verify_amd_endpoint.py means that process received HTTP 200. skipped_no_credentials is not a successful check.
- The default model for a new Stage-2 call is Gemma-4-31B-IT. The committed log above is Qwen3.8-Flash-Next, not Gemma. No committed Gemma latency.

7. AMD compute layer

- TRADEFLOW_SOLVER_BACKEND selects NumPy (default) or a CuPy/ROCm request.
- results/bench.json is a CPU measurement (is_gpu_active: false).
- The MI300X column is a theoretical roofline lower-bound estimate, not measured.
- Formula: lower_bound_seconds = (working_set_mb * 1024 * 1024) / (5.3e12).
- No AMD GPU measurement is in the repo unless a sibling benchmark JSON says ran_on_amd_gpu: true. Do not narrate a speedup.

8. Evidence, with the baseline named

- Self-play (results/self_play_benchmark.md): engine savings versus that table’s walk-away ceiling run from 8.8% to 20.0%. Hard-Bargainer 8.8% versus the fixed baseline 9.2%. 20 matches, 5 archetypes, seed 42. Not the auction headline.
- Real-model log, different experiment: Bluffing Hardliner deal rate 61.11%. 0 budget violations in 94 completed matches.
- Held-out intent accuracy on the synthetic set: 63.08% (195 of 500 held out).
- Auction headline is savings versus the first-round best quote. Versus target is shown, including when that figure is negative. The walk-away ceiling figure is secondary.
- Headline savings vs first-round best quote: 5.0% (CNY 6,600.00 total)
- Versus target: -5.0% (CNY -6,000.00 total). A negative figure means the settled price is above target.
- Secondary, versus walk-away ceiling: 22.2% (CNY 36,000.00 total)
- Offline comparison results/llm_vs_tradeflow.json: 24 episodes, seed 42, buyer offline_accept_opening. Not an AMD call. TradeFlow budget violations 0.00%, deal rate 50.00%, average savings versus the supplier opening 22.24%. Stand-in budget violations 100.00%, deal rate 100.00%, average savings 0.00%. Simulation. Not the auction headline.

9. Limits and roadmap

- cp .env.example .env && docker compose up --build, then open http://localhost:8000. render.yaml is the public-demo blueprint. No hostname is claimed until a deploy exists.
- Placeholder keys are ignored. /health reports offline_demo and the solver backend. AMD live mode stays off unless a real VLLM_API_KEY is set and the operator turns it on.
- No factory deal. No shipped MI300X timing. No committed Gemma latency. MIT. Public repo.
- Roadmap: keep the human on the send button, log a real Gemma chat only when a key is present, and publish extraction accuracy only after that log exists.
- Deadline per third-party listing: 18 Oct 2026 23:59 UTC (19 Oct 00:59 BST); on-site Luma pages list 16:00 local on 18 Oct; confirm on lablab.